AI ASSISTED CODING

ASSIGNMENT - 8

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Lab 8.2:

TASK-1:

PROMPT: Write a function is_even() generate test cases for finding if a number is even or odd

Code

```python
# Test cases
print(is_even(2))  # True
print(is_even(3))  # False
print(
print(
print(
print(is_even(10)) # True
print(is_even(11)) # False
print(is_even(100)) # True
#ASSERT TEST CASEES
assert is_even(2) == True
assert is_even(3) == False
assert is_even(0) == True
print("All assert test cases passed!")
#EXCEPTIONAL TEST CASE
try:
    is_even("a")
    assert False
except TypeError:
    print("Exception test passed")
#UNIT TESTCASES
import unittest
class TestIsEven(unittest.TestCase):
    def test_even(self):
        self.assertEqual(is_even(2), True)
        self.assertEqual(is_even(0), True)
        self.assertEqual(is_even(-4), True)
    def test_odd(self):
        self.assertEqual(is_even(3), False)
        self.assertEqual(is_even(11), False)
if __name__ == "__main__":
    unittest.main()
#PYTEST
def test_even_numbers():
    assert is_even(2) == True
    assert is_even(0) == True
def test_odd_numbers():
    assert is_even(3) == False
    assert is_even(11) == False
#DOC TEST
def is_even(num):
    """
    Returns True if the number is even, False otherwise.
```

Assignment-8.2.py > ...

```python
"""Write a function is_even() generate test cases for finding if a number is even or odd"""
def is_even(num):
    """Returns True if the number is even, False otherwise."""
    return num % 2 == 0
# Test cases
print(is_even(2))  # True
print(is_even(3))  # False
print(is_even(0))  # True
print(is_even(-4)) # True
print(is_even(-5)) # False
print(is_even(10)) # True
print(is_even(11)) # False
print(is_even(100)) # True
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
True
False
True
False
True
PS C:\Users\ARSHA THALLAPALLY\OneDrive\Desktop\AI-Assitent coding>
```

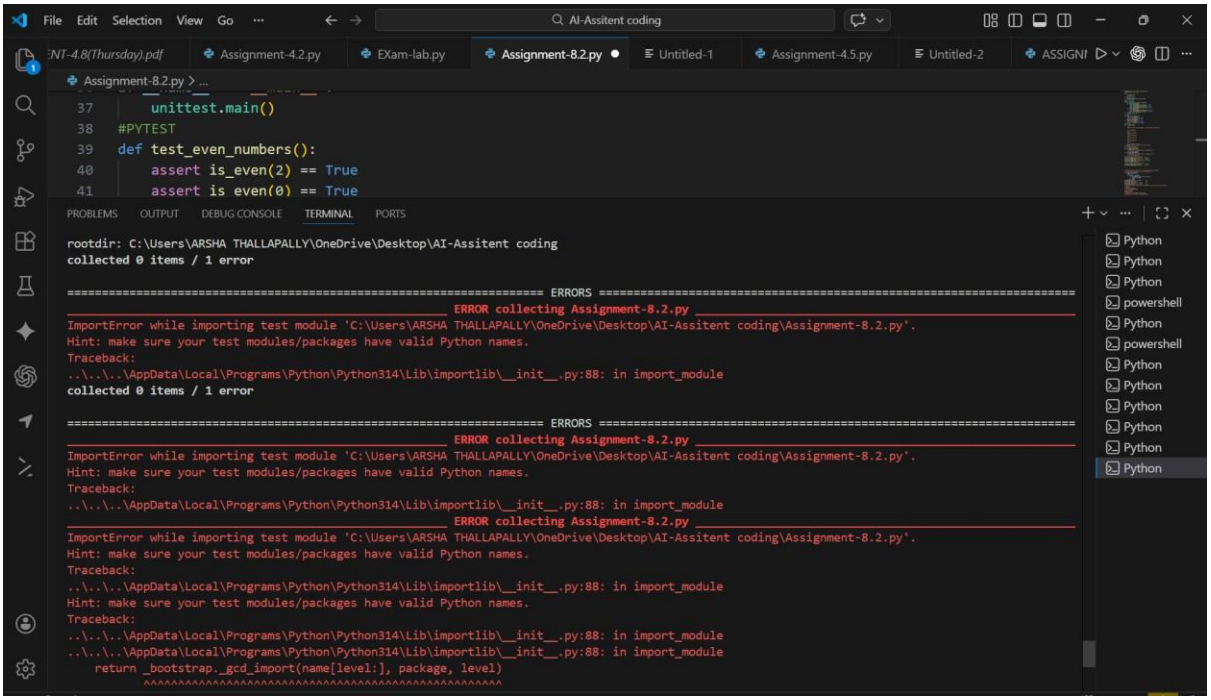
Assignment-8.2.py > ...

```python
"""Write a function is_even() generate test cases for finding if a number is even or odd"""
def is_even(num):
    """Returns True if the number is even, False otherwise."""
    return num % 2 == 0
# Test cases
print(is_even(2))  # True
print(is_even(3))  # False
print(is_even(0))  # True
print(is_even(-4)) # True
print(is_even(-5)) # False
print(is_even(10)) # True
print(is_even(11)) # False
print(is_even(100)) # True
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
True
False
True
False
True
PS C:\Users\ARSHA THALLAPALLY\OneDrive\Desktop\AI-Assitent coding>
```

PYTEST:



OBSERVATION :

1. Normal test case:

I printed the output of is_even() and checked manually if the numbers are even or odd.

2. Assert test case:

I used assert to automatically check if is_even() returned True for even numbers and False for odd numbers.

3. Exception test case:

I tested is_even() with invalid input like a string to see if it raises an error properly.

4. unittest test case:

I wrote a unittest class to test multiple even and odd numbers in an organized way.

5. pytest test case:

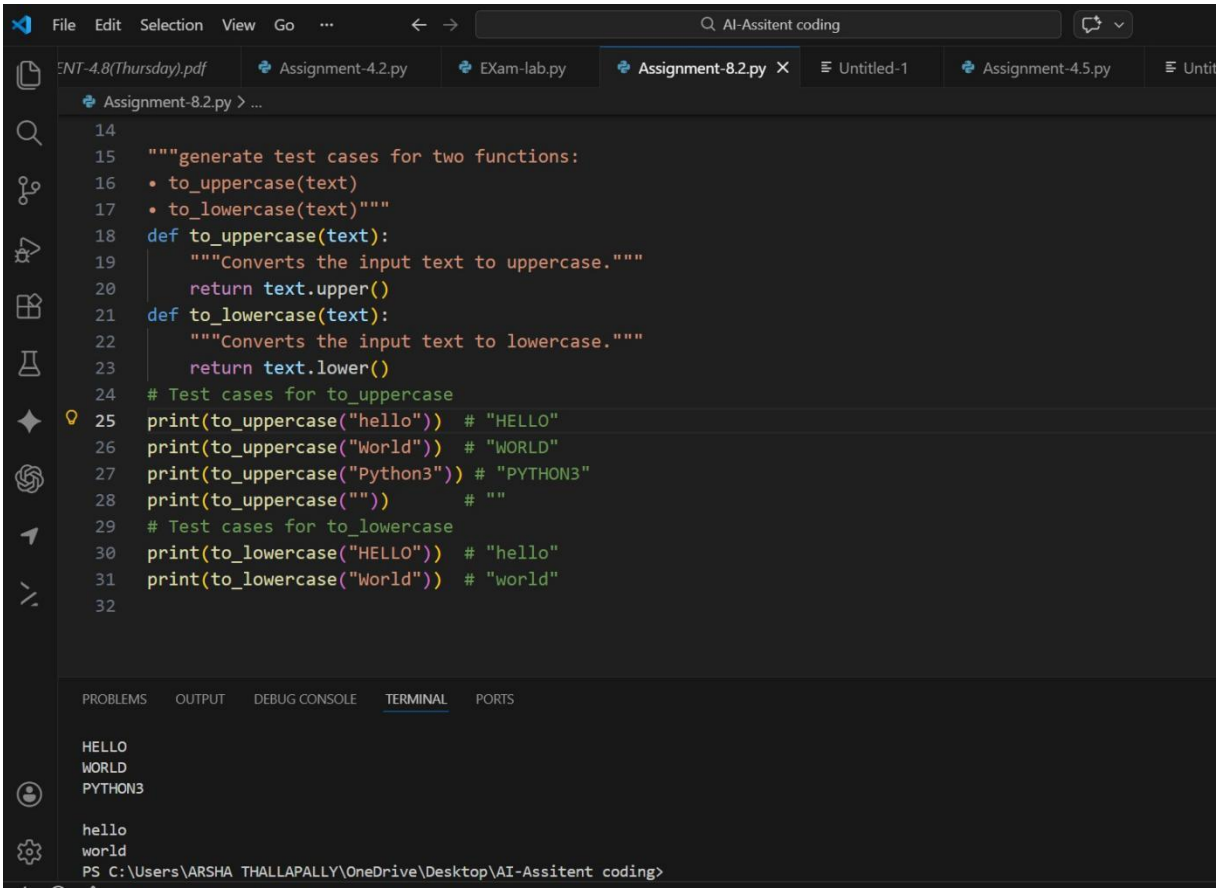
I used simple assert statements in pytest to check the function for different inputs.**Doctest:**

I added examples inside the function's docstring and ran doctest to automatically verify the outputs.

TASK-2

PROMPT: GENERATE TEST CASES FOR A FUNCTION SUM_LIST(NUMBERS)

THAT CALCULATES THE SUM OF LIST ELEMENTS CODE:



OBSERVATION:

The function correctly sums a list of positive numbers. For an empty list, it returns 0 as expected.

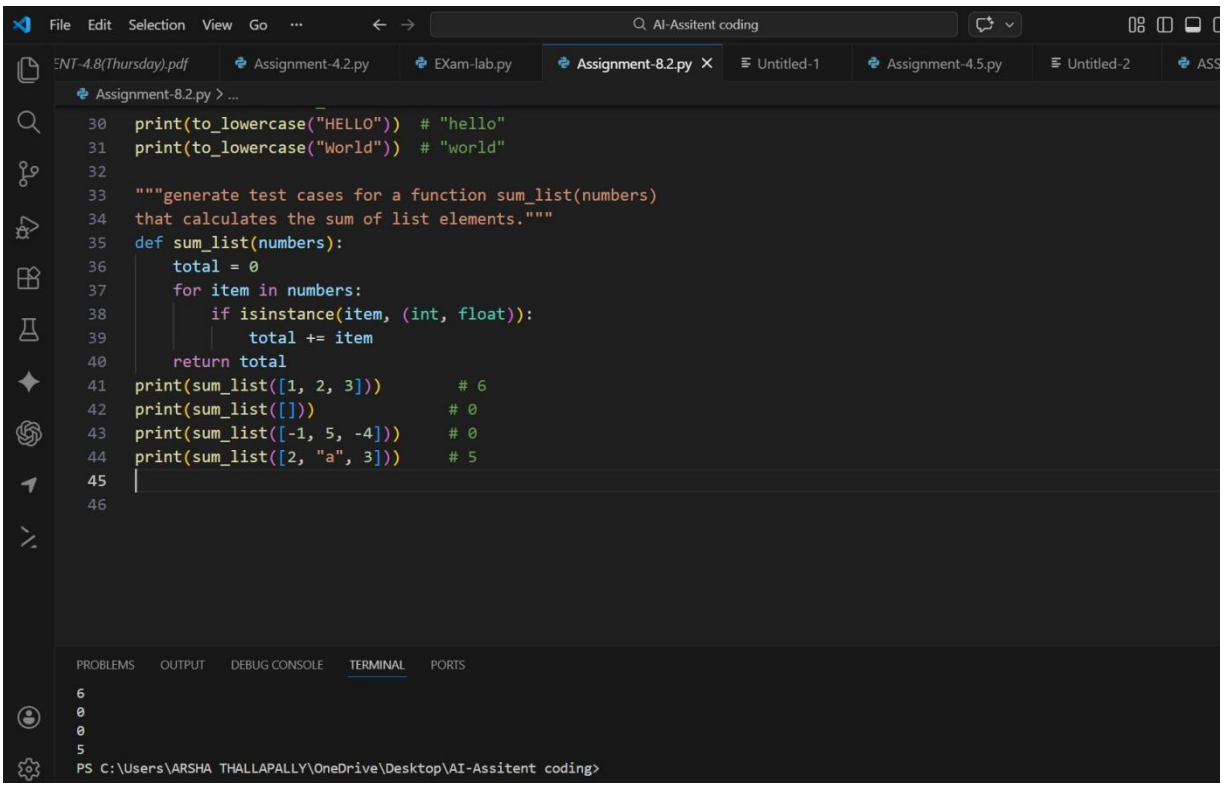
It handles a mix of negative and positive numbers properly

TASK-3

PROMPT:

generate test cases for a function sum_list(numbers) that calculates the sum of list elements.

CODE:



OBSERVATION:

Correctly calculates average and returns **Pass** for all marks above 40.

Returns **Fail** when average is below 40.

Passes at the boundary where average equals 40. Raises error for marks below 0.

Raises error for marks above 100.

TASK-4:

PROMPT:

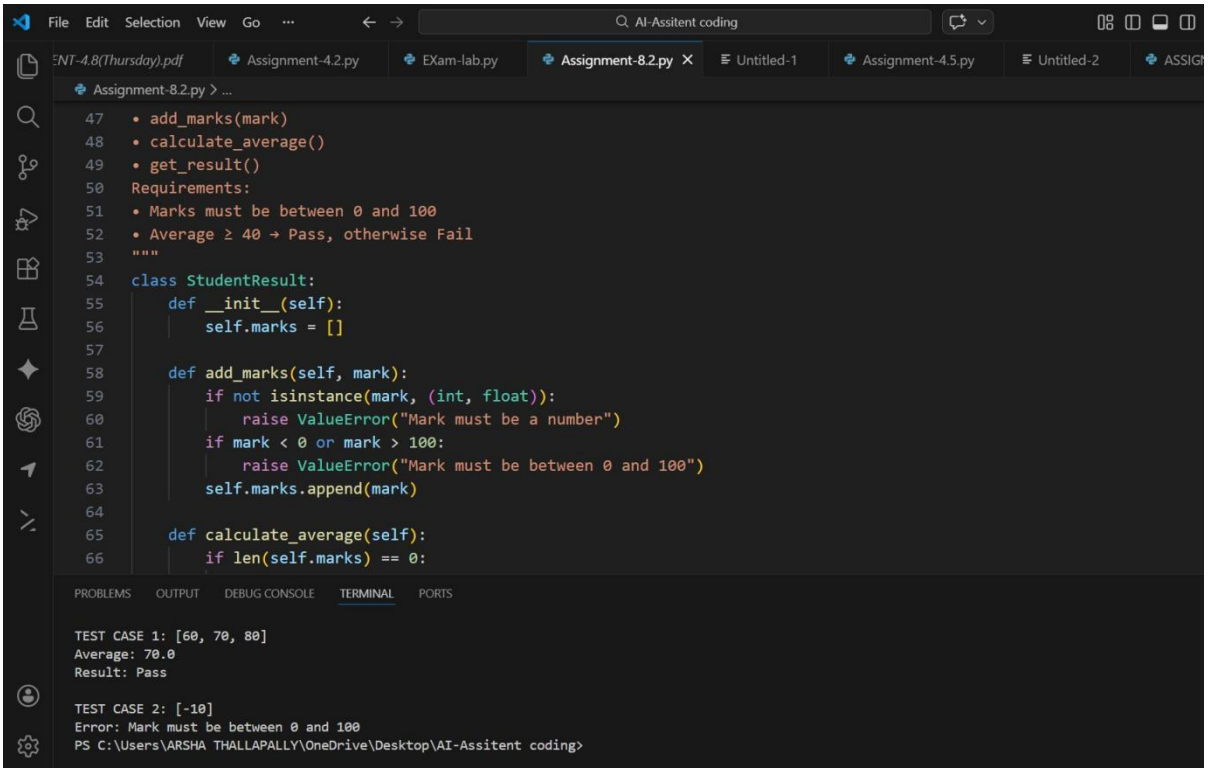
Generate test cases for a StudentResult class with the following methods:

- add_marks(mark)
- calculate_average()
- get_result()

Requirements:

- Marks must be between 0 and 100
- Average ≥ 40 → Pass, otherwise Fail

CODE:



TASK-5:

OBSERVATION:

The StudentResult class correctly calculates averages and returns Pass for averages ≥ 40 and Fail for averages < 40.

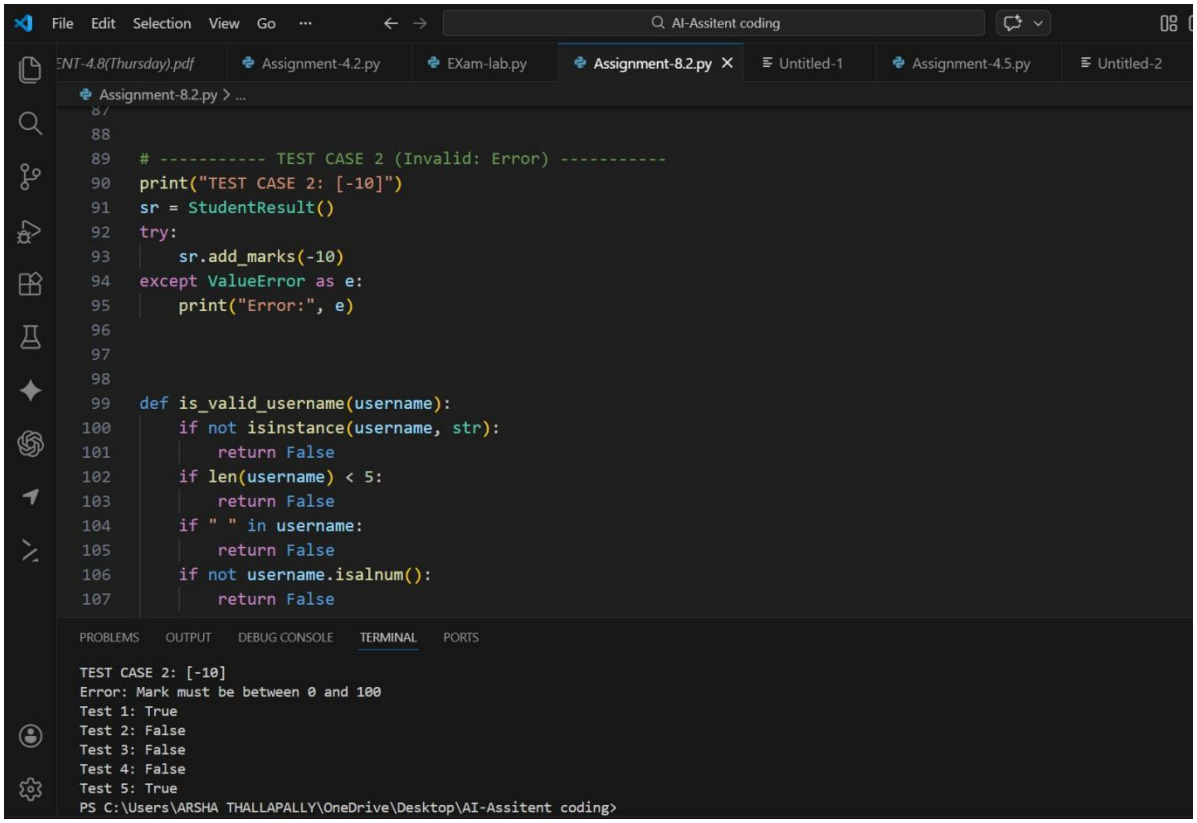
PROMPT:

It also raises errors for invalid marks (<0 or >100) or when no marks are added

TASK-5

PROMPT: Generate code for Minimum length: 5 characters,No spaces allowed, Only alphanumeric characters include all types of test cases mentioned

CODE:



OBSERVATION: The is_valid_username function correctly checks that usernames are at least 5 characters long, contain no spaces, and are alphanumeric.